

# The Autonomy Anchor Year: UK Maritime Autonomy Decision Intelligence Briefing, June 2026

Decision Intelligence · UK Maritime Autonomy · SAMPLE

## The Autonomy Anchor Year

24 June 2026 · Audience: NPL leadership, NSO leadership and the MAAT programme team · Horizon: 2 to 5 years

Read estimate: 1 min At a glance · 3 min Executive Synthesis · 15 min full read



**PREVIEW** · Sample deliverable for the proposed NPL / NSO continuous-intelligence programme. A live cycle would carry approximately 40 sources, full assurance audits and the Stehr Consulting domain authority overlay. The structure, source discipline and decision postures shown here are unchanged between sample and live cycle.

### What you will learn

1. Why 2026 is the year four UK and international clocks (Defence Investment Plan, MASS Code, MAAT, Crown Estate floating wind) converge on one decision window.
2. How the Stehr / NPL / NSO baseline of a £625m current footprint stacks against the demand and assurance evidence accumulated since 11 May 2026.
3. Which four implications turn the cycle's signals into Action Plan inputs, with a Decide / Prepare / Monitor posture on each.

### Key takeaways

1. IMO is on track to adopt a non-mandatory MASS Code in 2026, mandatory in 2028; the UK pathway must be visible to class and insurers before that lands.
2. The first Ghost Shark XL-AUV entered Royal Australian Navy service in February 2026, setting the allied production-rate benchmark the UK is now judged against.
3. OEUK quantifies the UKCS decommissioning workload at over £24bn through 2033; autonomous survey and inspection sit on the largest line item.

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THE AUTONOMY ANCHOR YEAR 1. Defence demand and the Hybrid Navy Immediate 2. Offshore energy + CUI protection Immediate 3. Assurance + certification pathway Near-term 4. Sovereign capability + cognitive layer Near-term 5. Capital, scale-up + export route Near-term 6. Workforce + regional clusters Longer-range Six themes, three tiers, one converging year

Six themes across three impact tiers. 2026 is the year the UK clocks line up.

## At a glance

UK maritime autonomy is being asked, in one calendar year, to graduate from a portfolio of pilots into a portfolio of platforms. Defence demand, offshore energy and undersea infrastructure protection all need it; assurance, capital and workforce have not yet caught up.

The IMO is on track to adopt the international rulebook for autonomous ships in 2026. The Royal Navy has committed to a hybrid crewed-uncrewed fleet. The Crown Estate sits on a 50 GW offshore wind pipeline. Class societies and the National Wealth Fund are deciding what to back. Allied navies are buying autonomy at production rate.

**The one thing to take into the next meeting:** the £3.7bn-by-2040 trajectory in the Stehr / NPL / NSO baseline depends less on whether UK autonomy works and more on whether the UK assurance pathway is visible to class, insurers and procurement before the Defence Investment Plan and the MASS Code both land. Track those two milestones as the single binding indicator.

## Executive Synthesis

Which procurement, investment and partnership decisions on your desk this quarter still assume the UK autonomy pathway is parallel to the international one, rather than racing it?

The IMO, the Royal Navy, the Crown Estate, the National Wealth Fund and three allied navies are all moving on the same 2026 to 2028 clock; the evidence in this report says the UK pathway must be visible to class and insurers before any of those clocks chime.

The cycle's central tension is timing. Six weeks after the Stehr / NPL / NSO study (SCL-082-007-R, 11 May 2026) set the baseline at a £625m current footprint and a £3.7bn-by-2040 central trajectory, the evidence accumulating around that baseline says the trajectory is now gated less by demand and more by the speed at which the UK assurance pathway becomes visible to the people who actually buy, insure and certify autonomous vessels. Three things have moved. The IMO has confirmed at MSC 109 that adoption of a non-mandatory MASS Code is on track for 2026 with mandatory entry under SOLAS in 2028 ([IMO](#), May 2026). The Royal Australian Navy has the first Ghost Shark XL-AUV in operational service, setting the production-rate benchmark allied navies are now measured against ([Royal Australian Navy](#), February 2026). And the Royal Navy mine-countermeasures transition is now operational rather than experimental, with the Hunt and Sandown classes on a published retirement path against an autonomous MCM portfolio ([Navy Lookout](#), April 2026).

The spine theme of the cycle is Theme 3, assurance and certification, because every other theme depends on it. Defence demand wants the kit (Theme 1). Offshore energy and CUI protection want the kit (Theme 2). Capital wants to back the kit (Theme 5). Allied navies want to buy the kit (also Theme 5). None of that converts to revenue at the rate the Stehr baseline assumes unless class, insurers and the MCA can certify the kit at the pace IMO is now pacing them at. The MGN 678 route published by the MCA in March 2025 is operating ([MCA](#), March 2025), Lloyd's Register's Unmanned Marine Systems Code is at revision 3 ([Lloyd's Register](#), September 2025), and DNV's class guideline DNV-CG-0264 sits alongside ([DNV](#), October 2025). What is missing is the visible UK confidence statement that ties MAAT outputs, MGN 678 routes and class notation into something a procurement officer can specify against.

The £3.7bn-by-2040 trajectory in the Stehr baseline depends less on whether UK autonomy works and more on whether the UK assurance pathway is visible to class, insurers and procurement before the Defence Investment Plan and the MASS Code both land.

Three further forces shape the cycle. **Defence pull:** the 2025 Strategic Defence Review ([MoD](#), June 2025) committed the Royal Navy to a hybrid crewed-uncrewed fleet; the Defence Investment Plan due in this calendar year will name the budget line. **Energy pull:** the Crown Estate's 50 GW by 2030 target ([Crown Estate](#), February 2025) and a UKCS decommissioning workload over £24bn through 2033 ([OEUK](#), November 2025) both sit on top of survey, inspection and intervention lines that autonomous platforms address. **CUI pull:** the JEF Nordic Warden activation after the December 2024 Eagle S incident ([GOV.UK](#), January 2025) and the standing NATO Critical Undersea Infrastructure Cell at MARCOM ([NATO](#), February 2024) have converted persistent autonomous surveillance from option to baseline assumption.

The reading worth surfacing for leadership: the sovereignty question is now a software question, not a hull question. The SeeByte Edinburgh team anchors NATO MCM autonomy from inside a US corporate parent ([Maritime Executive](#), April 2026); Sonardyne is expanding its in-house autonomy and acoustic-positioning software ([Splash 247](#), March 2026); the Stehr value-chain layer that needs domestic protection is the cognitive and information layer, not the platform layer. The Capital, scale-up and export route theme says the same thing from the financing side: ACUA Ocean closed a Series A in February 2026 ([Splash 247](#), February 2026), the National Wealth Fund mandate covers clean maritime ([National Wealth Fund](#), October 2025), and the Norwegian HUGIN replacement plus Australian SEA 5026 are the two allied procurement windows UK exporters will be tested against ([Maritime Executive](#), April 2026; [Australian Department of Defence](#), January 2026).

**Where this analysis could be wrong** The single most consequential assumption is that the Defence Investment Plan, when published, will name a distinct uncrewed-systems budget line at production rate rather than fold it inside the Type 26, Type 31 and FCASW envelopes. If the Plan publishes uncrewed as a sub-line of crewed programmes, the timing case for the UK assurance pathway weakens significantly: defence demand becomes lumpy, allied competitive pressure on UK class softens, and the Stehr conservative case (around £1bn footprint by 2040) becomes plausible. Evidence that would force a material revision: a Defence Investment Plan published with autonomous-systems funding folded into platform programmes, no named MCM autonomy line of effort, and a Royal Navy procurement update that defers the FCASW autonomy element to the 2030s.

### The decisions that cannot be deferred this cycle

1. **Publish a visible UK assurance pathway statement that ties MAAT outputs, MGN 678 routes and class notation together by Q4 2026.** The IMO MASS Code adoption window and the Defence Investment Plan publication window are now inside the same twelve-month period; a UK pathway statement after either lands is too late to shape the market.
2. **Treat the cognitive and information layer as the sovereign-capability priority, not the platform layer.** Software, mission systems and acoustic positioning are where the foreign-acquisition risk is concentrating and where UK SMEs are most thinly capitalised.
3. **Stand up the export sales architecture for Norway HUGIN replacement, Australia SEA 5026 and the Singapore MARSEC family of programmes by end of 2026.** These three windows are now in active procurement; the export case is a 2027 case if the route to market is not built this year.

Each is developed below, with a decision posture, in the four Strategic Implications.

### Audience Snapshots

Four lenses on the same intelligence base, one per audience type. Each card surfaces the one question this cycle puts to that audience.

Strategy & Programme Leadership

**Is the UK assurance pathway visible enough, early enough, to anchor the Action Plan?** **The shift:** IMO MASS Code adoption is on track for 2026; the Defence Investment Plan is due in the same calendar year. The MAAT programme's outputs become Action Plan inputs if and only if they are visible to class, insurers and procurement before either of those external clocks chimes.

**The question to brief:** Does the Shipbuilding & Maritime Technology Action Plan name MAAT outputs, MGN 678 routes and class notation as a single coherent UK assurance pathway, on a published timeline?

Industry & Supply Chain

**Have you sized the FY27 ROC, MCM and offshore-energy demand pools against your build rate?**

**The shift:** The Royal Navy MCM transition is operational, the Crown Estate 50 GW pipeline includes floating wind tranches, and the UKCS decommissioning workload through 2033 sits at over £24bn. Three demand pools, three distinct procurement clocks, all moving inside the cycle.

**The question to brief:** Which two of the five Stehr demand segments are you genuinely positioned to win in FY27, and where is your build-rate gap?

Assurance, Regulation & Standards

**Can UK assurance keep pace with class competitors through MSC 110, MSC 111 and the MASS Code adoption window?** **The shift:** Lloyd's Register, DNV and Bureau Veritas are all publishing class guidance on autonomous and remote-operations notation. The MGN 678 baseline is in place. The MAAT programme outputs are the missing piece tying the three together into a UK pathway that a procurement officer can specify against.

**The question to brief:** Is the MCA, MAAT, class and insurer interface formalised on a published cadence that matches the IMO clock?

Investors & Finance

**Are you backing the software layer at the rate the sovereignty question now demands?** **The shift:** ACUA Ocean closed a Series A; Sonardyne is growing its in-house autonomy software portfolio; SeeByte anchors NATO MCM autonomy from inside a US parent. The cognitive layer is where the sovereign-capability and the export-revenue arguments converge.

**The question to brief:** Does your maritime autonomy allocation explicitly carve out the software and mission-system layer as a sovereign-capability sleeve, or treat it as undifferentiated tech exposure?

## Themes

Six themes, in descending impact order. Each opens with the consequence for UK maritime autonomy, anchors three to four signals to dated sources, names the strongest counter-reading, and links to the Strategic Implication it shapes.

### 1. Defence demand and the Hybrid Navy as anchor market

Immediate

The Royal Navy is the closest thing the UK maritime autonomy sector has to a price-setting anchor customer; the Strategic Defence Review committed the service to a hybrid crewed-uncrewed fleet, and the mine-countermeasures transition is already operational. Demand is therefore not in doubt. What is in doubt is the pace at which procurement and assurance can convert that demand into clearable revenue: the binding constraint flagged by RUSI is the procurement-and-assurance clock, not the platform-supply side. Allied production rates (Australian Ghost Shark in operational service, US Replicator delivery cadence) now set the benchmark UK output is read against. The Defence Investment Plan, due this year, is the visibility signal that tells procurement and the supplier base whether the pace will accelerate; the budget-line nomenclature is one read on that, not the spine of the question.

**Figure 1. Allied autonomous-undersea production rate, selected programmes** Allied autonomous undersea / surface production rate, indicative Approximate annual planned production / fielding rates. Sources: US DoD Replicator, Royal Australian Navy Ghost Shark, Navy Lookout MHC coverage, Maritime Executive HUGIN replacement. US Replicator (small USV / UUV) ~1,000/yr Australia Ghost Shark (XL-AUV) ~12/yr scaling UK Royal Navy MHC portfolio ~20 platforms total Norway HUGIN replacement ~8/yr planned UK numbers are platform totals across the MHC programme; allied numbers are annual planned production rates.

Allied production rates set the benchmark the UK is now judged against. Comparing programme totals to annual rates flatters UK numbers; the underlying ratio is wider.

- **The SDR named uncrewed as a binding precondition, not an adjunct.** The Strategic Defence Review 2025 framed autonomous mine-countermeasures, unmanned underwater vehicles for ASW and a hybrid crewed-uncrewed surface fleet as load-bearing for Royal Navy mass through the 2030s ([MoD](#), June 2025).
- **The MCM transition is operational, not experimental.** The Mine Countermeasures and Hydrographic Capability programme has moved past pilot, with a published retirement path for the Hunt and Sandown classes and the SWEEP, SeaCat, Iver and RNMB Hazard autonomous portfolio in service rotation ([Navy Lookout](#), April 2026).
- **Allied production rate sets the benchmark.** The first Ghost Shark XL-AUV is in operational service with the Royal Australian Navy, with industrial scale-up under AUKUS Pillar 2 ([Royal Australian Navy](#), February 2026). The US Replicator initiative committed to thousands of attritable autonomous systems by August 2025 ([US Department of Defense](#), August 2024).
- **The bottleneck is procurement and assurance pace, not industry.** RUSI's recent reading is that the hybrid-fleet ambition is gated by the certification pathway for autonomous systems and the procurement model, not by industrial capacity ([RUSI](#), March 2026).

**Counter-argument** The strongest counter-reading is that the Defence Investment Plan, when published, will fold uncrewed funding inside the Type 26, Type 31 and FCASW envelopes rather than naming a separate production-rate line. If that happens, defence demand becomes a series of lumpy platform-tied opportunities rather than a stable pull on the SME base, and the Royal Navy's hybrid-fleet ambition becomes a 2030s case rather than a 2027 case. The MCM transition is the only clearly-funded operational autonomy line of effort; everything else sits inside platform programmes whose own schedules slip routinely.

**Decision link:** Strategic Implications [1](#) and [3](#).

## 2. Offshore energy and critical undersea infrastructure protection

Immediate

The UK has, in its own waters, two of the largest civil maritime autonomy demand pools in the world: a 50 GW offshore wind pipeline that needs persistent survey, inspection and maintenance, and a UKCS decommissioning workload over £24bn through 2033. To that has now been added a third, persistent surveillance of subsea cables and pipelines, which after the Eagle S incident and the JEF Nordic Warden activation can no longer be treated as a discretionary commitment. The consequence is that domestic civil and CUI demand is no longer the soft sister of defence demand: it is comparable in scale and runs on a faster commercial procurement clock.

- **The 50 GW pipeline carries floating wind tranches where autonomy economics are decisive.** The Crown Estate's Offshore Wind Report 2024 sets out the pipeline against the 50 GW by 2030 target, with floating wind the segment where autonomous survey, inspection and maintenance most directly determines the project economics ([Crown Estate](#), February 2025).
- **UKCS decommissioning workload is a durable, datable autonomy-addressable pool.** OEUK quantifies the UKCS decommissioning workload at over £24bn through 2033, with subsea, wells P&A and topside survey-and-inspection as the largest cost lines ([OEUK](#), November 2025).
- **CUI surveillance is now Alliance-level baseline, not exceptional commitment.** The JEF Nordic Warden activation after the December 2024 Eagle S incident in the Gulf of Finland ([GOV.UK](#), January 2025) and the standing NATO Critical Undersea Infrastructure Cell at MARCOM ([NATO](#), February 2024) sit on top of the same surveillance burden.
- **Crewed assets cannot carry the surveillance load alone.** The Atlantic Council's January 2026 issue brief on Europe's undersea cable defence reads the surveillance burden as exceeding crewed-asset capacity, with autonomous augmentation as a structural rather than optional measure ([Atlantic Council](#), January 2026).

**Counter-argument** The strongest counter-reading is that civil offshore-energy procurement is driven by short-cycle commercial tenders that European competitors (Saab, Kongsberg, Fugro-Norway) are already positioned to win, with assured class status and long-running operator relationships. Without a UK pathway statement that gives the same confidence to operators, the 50 GW pipeline and the decommissioning workload may be addressed by autonomous services, but bought from non-UK suppliers. On the CUI side, the JEF Nordic Warden activity may

consolidate around existing Defence Equipment & Support relationships rather than open as a fresh procurement window.

**Decision link:** Strategic Implications [1](#) and [4](#).

### 3. Assurance, regulation and the certification pathway

Near-Term

Assurance is the spine of this cycle because every other theme depends on it. The IMO is now publicly committed to adopting a non-mandatory MASS Code in 2026 with mandatory entry under SOLAS in 2028. The MCA has the MGN 678 baseline operating. Lloyd's Register, DNV and Bureau Veritas have published class guidance. The MAAT programme outputs are the missing tying element. The consequence is that the UK's competitive position over the next eighteen months is determined less by whether each of these pieces exists, and more by whether they are visible to procurement and insurance as one coherent pathway on a published cadence.

- **IMO MASS Code adoption is on a published clock.** MSC 109 confirmed the target for adoption of a non-mandatory MASS Code in 2026 and mandatory entry into force in 2028, with MSC 110 (June 2026) and MSC 111 the next decision points ([IMO](#), May 2026).
- **The MCA baseline is operating, not pending.** MGN 678 (March 2025) establishes the UK domestic regulatory route for MASS in UK waters, including registration, equipment standards and operator competence ([MCA](#), March 2025), and trial authorisations are being used at ABP Southampton and Forth Ports under that route ([Riviera Maritime Media](#), May 2026).
- **Class is paced, with UK and European competitors publishing in tandem.** Lloyd's Register's Unmanned Marine Systems Code reached revision 3 in September 2025 ([Lloyd's Register](#), September 2025); DNV's Class Guideline DNV-CG-0264 covers autonomous and remotely-operated ships on a risk-based basis ([DNV](#), October 2025).
- **The bottleneck is integration, not the individual pieces.** The MCA pathway, the MAAT outputs and the class notation routes function today as parallel tracks; procurement officers, underwriters and operators are asking for a single visible UK pathway statement rather than a triptych of separately-published artefacts.

**Counter-argument** The strongest counter-reading is that international class societies (DNV, BV) and the IMO MASS Code itself are converging fast enough that a separate UK pathway statement adds confusion rather than confidence. Operators and insurers may treat the IMO Code as the binding international rulebook and use class notation as the implementation route, with national regulators acting only on flag-state enforcement; in that world, a UK pathway statement is a domestic political artefact rather than a market-moving signal, and resources go into class engagement and IMO delegation work rather than into the pathway statement itself.

**Decision link:** Strategic Implications [1](#) and [2](#).

### 4. Sovereign capability and the cognitive layer

Near-Term

The Stehr value-chain layer that needs the most active sovereign-capability protection is the cognitive and information layer: autonomy software, mission systems, acoustic positioning and the cyber resilience of operational-technology stacks. The NSI Act regime is the binding filter, AUKUS Pillar 2 is the trilateral export channel, and NCSC plus IMO MSC.428(98) sit underneath as the cyber-assurance floor. The consequence is that sovereignty decisions over the next twelve months should treat software firms (SeeByte, Sonardyne software, BMT, AAC Clyde mission payloads) as the priority asset class to protect, not the hardware platforms.

- **The NSI Act regime now has a tracked pattern of intervention.** The DBT annual report on NSI Act interventions for 2024-25 sets out the volume and pattern of notifications, including defence and advanced-materials cases that map directly onto UK autonomy software and mission-system acquisition exposure ([DBT](#), September 2025).
- **AUKUS Pillar 2 maritime autonomy is the trilateral export channel.** The Cabinet Office update on AUKUS Pillar 2 advanced capabilities includes the maritime autonomy line of effort and the Maritime Big Play trilateral exercises that test interoperability of UUVs and USVs ([Cabinet Office](#), December 2025).

- **NCSC ships guidance is the OT cyber-assurance floor.** The NCSC Cyber Security for Ships collection establishes the OT cyber-resilience baseline, with autonomous and remotely-operated vessels carrying the largest OT attack surface ([NCSC](#), April 2025), aligning with IMO MSC.428(98).
- **UK software talent is anchoring NATO MCM autonomy from inside a US parent.** SeeByte's Neptune and Halcyon stacks deploy across NATO MCM fleets including the UK MHC programme, with the Edinburgh team the engineering anchor ([Maritime Executive](#), April 2026). Sonardyne is expanding its in-house autonomy and acoustic-positioning portfolio as a UK-owned counter-example ([Splash 247](#), March 2026).

**Counter-argument** The strongest counter-reading is that the cognitive layer is precisely the layer most efficiently delivered through allied partnership rather than sovereign duplication. AUKUS Pillar 2 maritime autonomy may be a stronger sovereign-capability protection than NSI Act enforcement against US acquirers, because it embeds UK software in the trilateral export channel rather than fencing it inside the UK procurement perimeter. On this reading, the SeeByte-inside-US-parent case is a feature, not a bug, and the sovereignty energy should go into trilateral data and IP-sharing standards under AUKUS rather than into widening NSI Act scope.

**Decision link:** Strategic Implications 2 and 3.

## 5. Capital, scale-up and the export route

Near-Term

The Stehr £8.3bn-by-2050 high case requires both a domestic capital channel that can fund SMEs from Series A through scale-up, and an export route to allied procurement. Both exist in fragmented form today. The consequence for the cycle is that the binding decision is sequencing: capital, export sales architecture and assurance need to be lined up so that an SME that raises a Series A in early 2026 has a credible Series B path on UK-mandate capital and a visible export pipeline by 2027.

- **The NWF (formerly UKIB) mandate covers clean maritime and ports.** The National Wealth Fund strategic priorities update names port infrastructure, clean maritime and supporting capabilities including autonomous vessel pilots as in-scope ([National Wealth Fund](#), October 2025); this is the long-duration UK capital channel for the layer that VC alone cannot reach.
- **Series A activity is live in the clean-propulsion-plus-autonomy intersection.** ACUA Ocean closed a Series A in February 2026 for the HUVAR-class hydrogen-fuelled autonomous surface vessel ([Splash 247](#), February 2026), an early proof point that growth capital is reachable for the dual-mandate platform.
- **Three allied procurement windows are in active phase.** The Royal Norwegian Navy HUGIN replacement is the largest near-term allied AUV procurement ([Maritime Executive](#), April 2026); Australian Project SEA 5026 is the AUKUS-relevant autonomous MCM and hydrographic window ([Australian Department of Defence](#), January 2026); the Singapore MARSEC family of programmes is the third.
- **Industry-body scaffolding is in place.** Maritime UK's State of the Maritime Nation 2025 sets out the cross-cluster industry voice on export ambition, regional clusters and the maritime-tech finance gap ([Maritime UK](#), July 2025), the umbrella to which DBT export campaigns can hook.

**Counter-argument** The strongest counter-reading is that allied procurement programmes route through their own national prime contractors and that UK firms compete only at the sub-system tier. Kongsberg holds the HUGIN incumbency for Norway; Anduril Australia is the Ghost Shark prime; Saab and Atlas Elektronik are the European MCM incumbents. UK firms can win at the sensor, software or mission-payload layer but the platform prime layer is not addressable at scale. On that reading, the £8.3bn-by-2050 high case overstates the share of allied procurement value UK firms can capture, and the realistic case is closer to the Stehr central case.

**Decision link:** Strategic Implications 2 and 4.

## 6. Workforce, regional clusters and the operating-model shift

Longer-Range

The longer-range structural factor is that the operating-model shift to remote-operations-centre (ROC) crewing changes the geography, the workforce profile and the public-trust posture of UK maritime. Regional clusters at Plymouth (Smart Sound, DDoP), the Solent, Aberdeen and Tyneside become the physical anchors. The seafarer

pipeline becomes a binding constraint. The CHELSEA and OceanGate-era lessons frame the public-trust task. The consequence for the cycle is that the workforce and clusters question is the one most likely to be under-prioritised today and to become decisive in 2028 to 2030.

### Weak signals to track

- Weak Royal Navy ROC operator career pathway prototypes emerging from the Sonardyne and MSubs training pipelines: gains weight if Royal Navy Career 2050 commentary names a distinct ROC operator branch.
- Weak Plymouth and Tyneside regional clusters starting to compete for the same talent pool: gains weight if a high-profile autonomy SME relocates between clusters or signs a multi-year talent partnership with a Catapult.
- Weak Insurance underwriting behaviour after the first contested autonomous-vessel near-miss or incident in UK waters: gains weight if IUMI publishes guidance or Marsh moves on standard autonomous-vessel cover language.
- **The seafarer pipeline is the binding workforce constraint.** The Maritime Skills Commission UK Seafarer Projections 2024 quantify the pipeline shortfall against which any ROC crewing model must be planned ([Maritime Skills Commission](#), November 2024).
- **Plymouth is the leading UK regional anchor.** Smart Sound Plymouth's advanced marine technology testbed and the DDoP defence cluster anchor marine autonomy testing, training and supply-chain co-location ([Smart Sound Plymouth](#), September 2025); the Solent, Aberdeen and Tyneside complete the multi-cluster map ([Connected Places Catapult](#), June 2025).
- **The MAIB casework frames the public-trust task.** The Marine Accident Investigation Branch Annual Report 2024 documents the human-factors and crewing failures that inform the public-trust posture for ROC crewing ([MAIB](#), July 2025); the CHELSEA and OceanGate-era public-trust lessons sit underneath as the cautionary frame.

**Counter-argument** The strongest counter-reading is that the operating-model shift to ROC crewing is overstated for the late 2020s. Crewed-vessel demand from cabotage, offshore wind installation and complex offshore operations remains robust; ROC crewing is realistic for survey, MCM and persistent surveillance but not for the bulk of UK maritime work through 2030. On this reading, workforce planning should still be sized for crewed-vessel growth, and the regional-cluster question is a matter of where the autonomy training overlay sits, not a wholesale operating-model migration.

**Decision link:** Strategic Implication 4.

## Strategic Implications

Four decisions turn this cycle's signals into Action Plan inputs. Each names the move, a horizon and a decision posture.

**SI 1: Publish a single visible UK assurance pathway statement by Q4 2026** NPL (via MAAT), the MCA and Lloyd's Register should jointly publish, by Q4 2026, a single UK assurance pathway statement that ties MAAT outputs, MGN 678 routes and class notation together with a published cadence aligned to the IMO MASS Code adoption window. The statement should be specific enough that a procurement officer can specify against it and an underwriter can price it. The reason for Q4 2026 is that the IMO MASS Code adoption window and the Defence Investment Plan window both fall inside the following twelve months; a pathway statement after either lands is too late to shape the market.

Action: NPL MAAT, MCA and Lloyd's Register to publish a joint UK MASS Assurance Pathway statement by 30 November 2026, with named milestones through to MSC 111.

Decide Draws on Themes 1, 2 and 3.

**SI 2: Carve the cognitive and information layer out as a distinct sovereign-capability sleeve** The NSO, working with the Investment Security Unit and the National Wealth Fund, should formally treat autonomy software, mission-system and acoustic-positioning firms as a distinct sovereign-capability sleeve rather than absorbing them into general defence-tech or maritime-tech exposure. The reason is that the SeeByte case demonstrates UK software talent can anchor NATO MCM autonomy from inside a US parent; the question for sovereignty is not whether the

platform is built here, but whether the cognitive layer is owned or partnered on terms the UK can renegotiate. Prepare posture rather than Decide because the precise instrument (revised NSI scope, NWF sleeve, AUKUS Pillar 2 IP-sharing standard) needs work before commitment.

Action: NSO to commission a cognitive-layer sovereign-capability scoping paper for ministerial consideration by Q1 2027, against the NSI Act, NWF and AUKUS Pillar 2 instruments.

Prepare Draws on Themes 3, 4 and 5.

**SI 3: Stand up export sales architecture for Norway, Australia and Singapore by end of 2026** DBT, the NSO and the relevant primes should stand up, by end of 2026, a coordinated export sales architecture for the three live allied procurement windows: the Royal Norwegian Navy HUGIN replacement, Australian Project SEA 5026 and the Singapore MARSEC family of programmes. The reason is that all three are in active procurement phase, and UK firms that arrive without a coordinated route to market in 2026 will be relegated to sub-system tier work in 2027. Prepare rather than Decide because the architecture (consortium model, IP carve-outs, AUKUS Pillar 2 routing) requires negotiation with industry and with allied partners.

Action: NSO and DBT to convene a UK Maritime Autonomy Export Committee with industry by 30 September 2026, with named pursuits in Norway, Australia and Singapore.

Prepare Draws on Themes 1, 4 and 5.

**SI 4: Monitor the ROC operator workforce question for a 2028 decision point** The NSO and Maritime Skills Commission should track the emerging ROC operator workforce question through 2027, with a formal decision point in 2028 on whether the UK needs a dedicated ROC operator training and licensing route distinct from existing seafarer certification. The reason is that the workforce question is a longer-range structural factor: the seafarer pipeline shortfall is real, the operating-model shift to ROC crewing is plausible but uneven, and the public-trust lessons from MAIB casework and the CHELSEA / OceanGate-era frame the politics. Monitor posture because the evidence base is not yet sufficient to fix a UK position, and class plus the IMO MASS Code will help shape the international competence baseline first.

Action: Maritime Skills Commission to publish a ROC operator workforce baseline by Q2 2027, against which a 2028 decision can be taken.

Monitor Draws on Theme 6.

## Scenario Matrix

The two critical uncertainties shaping UK maritime autonomy 2 to 5 years out are (vertical axis) the pace at which the UK assurance and regulatory pathway matures, fast at top and slow at bottom, and (horizontal axis) domestic demand pull across defence, offshore energy and CUI procurement, weak on the left and strong on the right. The four scenarios below are planning aids, not forecasts.

Domestic demand pull (defence, offshore, CUI)

UK assurance pathway pace

Fast · Weak

Fast · Strong

Slow · Weak

Slow · Strong

**Test-Bed Britain** UK class and regulatory pathway publishes ahead of the IMO MASS Code; the MCA, MAAT and Lloyd's Register integrate cleanly; UK becomes the certifier-of-choice for allied autonomous vessels. But the Defence Investment Plan folds uncrewed inside platform programmes, the Crown Estate operators specify allied incumbents, and CUI procurement consolidates around DE&S relationships. UK SMEs export the certified product but the design and integration value-chain layers concentrate offshore in Norway, the US and France. The Stehr footprint trajectory holds at the central case but the value capture is thin.

**Early indicators:** Joint MCA, MAAT and class pathway statement published Q4 2026; allied operators citing UK class for non-UK-built vessels; Defence Investment Plan with no named uncrewed budget line.

**Globally-Anchored Growth** UK assurance pathway publishes on a visible cadence; the Defence Investment Plan names a distinct uncrewed line of effort; the Crown Estate floating wind tranches specify autonomous survey and inspection routes; CUI procurement opens beyond DE&S relationships. UK firms win at platform prime, integrator and software layers across MHC, the Norwegian HUGIN replacement and Australian SEA 5026. NWF, NSSIF and private growth capital scale Series B and C rounds. This is the Stehr high case territory and the £8.3bn-by-2050 trajectory becomes credible.

**Early indicators:** Pathway statement plus named Defence Investment Plan uncrewed line by Q4 2026; ACUA-class follow-on Series B rounds in 2027; UK firm winning prime role in at least one allied programme.

**Value Leakage** UK assurance pathway slips behind the IMO clock; class output from Lloyd's Register lags DNV and BV; UK SMEs cannot prove a credible domestic certification route to allied buyers. Capital migrates: NSSIF, growth equity and private credit price the sovereignty risk in but at a deep discount. SeeByte-pattern acquisitions accelerate; ACUA Ocean and Marine AI become acquisition targets rather than scale-up cases. The Stehr conservative case prevails: footprint plateaus near £1bn by 2040, value capture is largely sub-system.

**Early indicators:** Pathway statement slipping into 2027; an NSI Act intervention publicly declined or reversed; UK autonomy SME relocating headquarters to the US or Norway.

**Demand-Pull Without Plumbing** Defence Investment Plan funds uncrewed at production rate; offshore wind floating tranches name autonomous services as required; CUI procurement opens up. But UK assurance is not visible enough for class, insurers and procurement to specify UK firms; underwriters price UK-flagged autonomous operation at a premium; imports fill the demand. UK industrial base struggles to convert demand into revenue at the rate the demand would otherwise allow. The Stehr central case holds but the import share is high; the export case suffers because UK firms cannot point to a domestic certified product.

**Early indicators:** Defence Investment Plan with uncrewed line of effort but no parallel pathway statement; rising allied imports in MHC and offshore survey segments; Marsh or IUMI guidance treating UK autonomous cover as elevated risk.

## What We Are Not Planning For

**An IMO MASS Code adoption failure at MSC 110 or MSC 111** The IMO MASS Code workstream is on a published clock that has moved consistently through 2024 and 2025 and now into 2026. The MSC 109 outcomes reaffirmed the 2026 adoption target. A failure at MSC 110 or MSC 111 is possible but not the planning baseline; the planning assumption is adoption on track with the customary refinements rather than collapse.

Reinstate if: MSC 110 outcomes (June 2026) defer adoption to 2027 or later, or an IMO Member State publicly withdraws support for the MASS Code timeline.

**A wholesale UK pivot away from AUKUS Pillar 2 maritime autonomy** AUKUS Pillar 2 maritime autonomy is the trilateral export channel that UK sovereign-capability calculations now sit on top of. A change of US administration or an Australian political shift could change the rhythm of cooperation but the trilateral architecture is durable enough that a wholesale UK pivot away is not the planning baseline. The planning assumption is continuation with periodic recalibration.

Reinstate if: an AUKUS partner formally suspends Pillar 2 maritime autonomy cooperation, or a UK government publicly de-prioritises the trilateral track for the Atlantic Alliance or a bilateral.

**A near-term collapse in offshore wind capex against the 50 GW target** The Crown Estate 50 GW by 2030 target carries known cost-pressure risks and several Allocation Round outcomes have been weaker than industry expected. A wholesale collapse of the pipeline is possible but the floating wind tranches and the operations and maintenance demand from already-consented capacity will not vanish; the autonomy demand from inspection, survey and maintenance is more durable than the Allocation Round headline numbers suggest.

Reinstate if: a major UK offshore wind developer cancels two or more consented projects, or the UK government revises the 50 GW by 2030 target downward by more than 25 percent.

### Discussion Points for the Leadership Team

1. If the joint UK assurance pathway statement does not publish by Q4 2026, which Stehr demand segment (defence, offshore energy, CUI, ports, deep-sea commercial) takes the largest near-term hit, and how does that change the FY27 prioritisation of MAAT outputs?
2. Where is the line between sovereign-capability protection of the cognitive layer and the trilateral interoperability AUKUS Pillar 2 needs? Specifically, is the SeeByte-inside-US-parent pattern a feature or a bug, and what would a UK government answer to that question look like in a published policy line?
3. Of the three allied procurement windows (Norway HUGIN replacement, Australia SEA 5026, Singapore MARSEC), which one would the cycle prioritise if UK industry capacity to pursue all three at full intensity is not realistic, and what does that say about the export sales architecture’s design?
4. What public-trust posture should the NSO take in advance of the first contested autonomous-vessel near-miss or incident in UK waters, drawing on the MAIB casework and the CHELSEA and OceanGate-era public-trust lessons rather than waiting for the event itself?
5. Drawing on the weak signal in Theme 6: if Plymouth (Smart Sound, DDoP) and Tyneside (Catapult) start competing for the same autonomy talent pool in 2027, does the NSO have a position on regional cluster prioritisation, or is the policy line that competition between clusters is healthy for the sector?

### Source Confidence Register

This briefing draws on 30 verified sources, gathered under a soft six-month recency window (publications from December 2025 onward), with 14 structural anchors carried as longer-running legislation, programme baselines and regulatory routes that the cycle’s analysis depends on.

Source tiers: **Tier 1**, governments, regulators and intergovernmental bodies. **Tier 2**, think-tanks, academic institutes, major consultancies and quality data providers. **Tier 3**, quality journalism and specialist trade press. **Tier 4**, vendor, company and practitioner sources, used only as directional corroboration.

#### Theme 1: Defence demand and the Hybrid Navy as anchor market

| Source                                                    | Tier   | Date     | Key claim used                                            |
|-----------------------------------------------------------|--------|----------|-----------------------------------------------------------|
| <a href="#">UK MoD: Strategic Defence Review 2025</a>     | Tier 1 | Jun 2025 | Frames autonomous MCM, UUV-ASW and a hybrid cre           |
| <a href="#">RUSI: The Royal Navy’s Hybrid Future</a>      | Tier 2 | Mar 2026 | Reads the binding constraint as procurement and assura    |
| <a href="#">Navy Lookout: RN MCM autonomy transition</a>  | Tier 3 | Apr 2026 | Documents the MHC operational progression and Hunt a      |
| <a href="#">US DoD: Replicator Initiative</a>             | Tier 1 | Aug 2024 | Provides the allied pacing benchmark on attritable auton  |
| <a href="#">Royal Australian Navy: Ghost Shark XL-AUV</a> | Tier 1 | Feb 2026 | First Ghost Shark in operational service, with industrial |

#### Theme 2: Offshore energy and critical undersea infrastructure protection

| Source                                                                    | Tier   | Date     | Key claim used                                |
|---------------------------------------------------------------------------|--------|----------|-----------------------------------------------|
| <a href="#">OEUK: Decommissioning Insight 2025</a>                        | Tier 1 | Nov 2025 | UKCS decommissioning workload over £24bn      |
| <a href="#">The Crown Estate: Offshore Wind Report 2024</a>               | Tier 1 | Feb 2025 | Sets out the pipeline against the 50 GW by 2  |
| <a href="#">NATO: Critical Undersea Infrastructure Cell</a>               | Tier 1 | Feb 2024 | Persistent autonomous surveillance is the eco |
| <a href="#">UK Govt: JEF Nordic Warden activation</a>                     | Tier 1 | Jan 2025 | JEF tracking shadow-fleet vessels near subsea |
| <a href="#">Atlantic Council: Defending Europe’s cables and pipelines</a> | Tier 2 | Jan 2026 | Surveillance burden exceeds crewed-asset cap  |

#### Theme 3: Assurance, regulation and the certification pathway

| Source                                    | Tier   | Date     | Key claim used                                 |
|-------------------------------------------|--------|----------|------------------------------------------------|
| <a href="#">IMO: MASS Code workstream</a> | Tier 1 | May 2026 | Adoption of non-mandatory MASS Code on track f |

| Source                                                           | Tier   | Date     | Key claim used                                     |
|------------------------------------------------------------------|--------|----------|----------------------------------------------------|
| <a href="#">MCA: MGN 678 UK domestic MASS regulation</a>         | Tier 1 | Mar 2025 | UK domestic regulatory baseline for MASS in UK v   |
| <a href="#">Lloyd's Register: Unmanned Marine Systems Code</a>   | Tier 1 | Sep 2025 | UMS Code rev 3, SHIPRIGHT design assessment a      |
| <a href="#">DNV: Class Guideline DNV-CG-0264</a>                 | Tier 2 | Oct 2025 | Risk-based class guideline for autonomous and remo |
| <a href="#">Riviera Maritime Media: UK ports autonomy trials</a> | Tier 3 | May 2026 | ABP Southampton and Forth Ports trials use MGN     |

#### Theme 4: Sovereign capability and the cognitive layer

| Source                                                      | Tier   | Date     | Key claim used                                        |
|-------------------------------------------------------------|--------|----------|-------------------------------------------------------|
| <a href="#">UK DBT: NSI Act annual report 2024-25</a>       | Tier 1 | Sep 2025 | Volume and pattern of NSI Act interventions includin  |
| <a href="#">UK Cabinet Office: AUKUS Pillar 2 update</a>    | Tier 1 | Dec 2025 | AUKUS Pillar 2 maritime autonomy line of effort and   |
| <a href="#">NCSC: Cyber Security for Ships</a>              | Tier 1 | Apr 2025 | OT cyber-resilience baseline for shipboard systems al |
| <a href="#">Maritime Executive: SeeByte across NATO MCM</a> | Tier 3 | Apr 2026 | SeeByte Edinburgh team anchors NATO MCM auton         |
| <a href="#">Splash 247: Sonardyne autonomy software</a>     | Tier 3 | Mar 2026 | Sonardyne expanding in-house autonomy and acousti     |

#### Theme 5: Capital, scale-up and the export route

| Source                                                         | Tier   | Date     | Key claim used                                        |
|----------------------------------------------------------------|--------|----------|-------------------------------------------------------|
| <a href="#">National Wealth Fund: Strategic priorities</a>     | Tier 2 | Oct 2025 | NWF mandate covers port infrastructure, clean mari    |
| <a href="#">Splash 247: ACUA Ocean Series A</a>                | Tier 3 | Feb 2026 | Series A round for hydrogen-fuelled HUVAR-class au    |
| <a href="#">Maritime Executive: Norway HUGIN replacement</a>   | Tier 3 | Apr 2026 | Royal Norwegian Navy HUGIN replacement is the la      |
| <a href="#">Australian DoD: Project SEA 5026</a>               | Tier 1 | Jan 2026 | Australian SEA 5026 autonomous MCM and hydrogr        |
| <a href="#">Maritime UK: State of the Maritime Nation 2025</a> | Tier 2 | Jul 2025 | Industry-body articulation of export ambition, cluste |
| <a href="#">gCaptain: China autonomous LNG carrier pilot</a>   | Tier 3 | May 2026 | First Chinese fully-autonomous LNG carrier pilot; no  |

#### Theme 6: Workforce, regional clusters and the operating-model shift

| Source                                                                     | Tier   | Date     | Key claim used                           |
|----------------------------------------------------------------------------|--------|----------|------------------------------------------|
| <a href="#">Maritime Skills Commission: UK Seafarer Projections 2024</a>   | Tier 2 | Nov 2024 | UK seafarer pipeline gap; binding constr |
| <a href="#">Smart Sound Plymouth: testbed and DDoP cluster</a>             | Tier 1 | Sep 2025 | Plymouth marine autonomy testbed and     |
| <a href="#">Connected Places Catapult: Maritime Innovation Cluster Map</a> | Tier 2 | Jun 2025 | Maps UK maritime innovation cluster lar  |
| <a href="#">MAIB: Annual Report 2024</a>                                   | Tier 1 | Jul 2025 | MAIB casework on human-factors and cr    |

### Claim-fidelity self-disclosure

This appendix records every claim that goes beyond what a cited source literally states, so the reader can see where analyst inference does the work and where the source carries it.

**Cycle-over-cycle continuity claims** This is the inaugural cycle for the UK Maritime Autonomy sample. The only cycle-over-cycle anchor is the Stehr / NPL / NSO study (SCL-082-007-R, 11 May 2026), which is treated as the cycle's external baseline rather than a prior in-series memory entry. References to the £625m current footprint, the £3.7bn-by-2040 central trajectory, the £8.3bn-by-2050 high case and the value-chain layer language trace to that single external baseline; subsequent cycles will trace continuity through this report's `memory.json` entry.

**Superlatives used** *"Largest near-term allied AUV procurement"* for the Norwegian HUGIN replacement (Theme 5): analyst characterisation supported by the Maritime Executive April 2026 framing, but not a direct verbatim claim from that source. *"Leading UK regional anchor"* for Plymouth (Theme 6): analyst characterisation against the Connected Places Catapult cluster map. *"Load-bearing"* applied to the Strategic Defence Review's uncrewed framing in the Executive Synthesis: analyst rendering of the SDR text, which uses "essential" and "central" language but not the literal phrase "load-bearing".

**Analytical synthesis claims** The central claim that the spine of the cycle is Theme 3 (assurance) draws on the convergence of the IMO MASS Code timeline, the MCA MGN 678 baseline, the LR and DNV class guidelines and the Defence Investment Plan publication window; no single source makes the assurance-as-spine argument, and the reading is the analyst's. Taken together, the sources suggest that the binding constraint on the Stehr trajectory is the visibility of the UK pathway rather than any one of its components. The claim that the sovereignty question is now a software question, not a hull question, is similarly an analyst synthesis across the SeeByte, Sonardyne, NSI Act and AUKUS Pillar 2 evidence; no single source makes the claim in those words.

This Decision Intelligence briefing is the quarterly synthesis of a Shaping Tomorrow intelligence cycle on **UK Maritime Autonomy**. It sits on top of weekly Signal Scans that identify emerging developments and weak signals, and monthly Change Trackers that assess what is materially shifting in momentum. Cycles may also close with Action Triggers, an end-of-cycle synthesis that reads across all four formats and surfaces patterns no single deliverable carries on its own. Continuous intelligence on this topic is available as a Single-Topic or Multi-Topic programme.

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